AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

Geometric and Performance Analysis of Recursive Multi-Agent Systems for Enterprise Process Automation

Team D01:

Harshvardhan V – CB.SC.U4AIE23353

Rishi Raajha A – CB.SC.U4AIE23335

Naveen Sankar R S – CB.SC.U4AIE23323

Sakthi Pranav S– CB.SC.U4AIE23336

Supervisor:

Dr. Prem Jagadeesan

Introduction

- Enterprise workflows (procurement, order fulfillment, customer support, and IT service management) involve complex, interconnected tasks requiring planning, tool usage, and decision-making.
- LLM-based multi-agent systems have emerged as a promising solution for automating these enterprise processes.
- Recursive Multi-Agent Systems (RecursiveMAS) improve collaboration by enabling agents to iteratively exchange latent representations through recursive communication, enhancing reasoning and task execution
- As workflow complexity and agent interactions increase, system reliability and task performance can degrade.
- Evaluating both performance metrics and the geometric behavior of agent representations can provide deeper insights into system reliability.

Literature Review

Paper	Methodology	Dataset	Results	Limitations
Recursive Multi-Agent Systems(Zou et al., 2026)	Trains a lightweight connector module (RecursiveLink) that enables LLM agents to exchange latent states recursively	Evaluated on 9 benchmark datasets, including MATH500, AIME2025/2026, GPQA-Diamond, MedQA, LiveCodeBench, and HotpotQA	Improved accuracy by 8.3%, achieved up to 2.4× faster inference, and reduced token usage by up to 75.6%	Lacks enterprise workflow evaluation and detailed geometric representation analysis.
Neural Representational Geometry of Concepts in Large Language Models (Li et al., 2024)	Uses geometric manifold analysis with SNR, dimensionality, bias, overlap, and PCA to study latent representations across LLM layers.	LLaMA-3-8B and OpenAI text-embedding-3-large using 25 concept categories with generated sentence embeddings.	Shows that geometric properties accurately explain few-shot classification performance and reveals how latent representations evolve across layers.	Focuses on single LLM concept representations and does not analyze recursive multi-agent systems, enterprise workflows, or collaborative latent representations.

Research Gaps

- RecursiveMAS is evaluated on generic reasoning benchmarks but not on enterprise process automation workflows, limiting its applicability to real-world enterprise scenarios.
- Existing work on neural representational geometry analyzes latent representations of single LLMs using geometric manifold analysis, but does not investigate recursive multi-agent systems or collaborative latent-state interactions.
- No existing work integrates recursive multi-agent architectures with comprehensive geometric analysis of latent representations to evaluate system reliability as enterprise workflow complexity increases.

Problem Statement

- Enterprise workflows such as procurement, order fulfillment, customer support, and IT service management involve complex tasks requiring planning, tool invocation, decision-making, and verification. Although LLM-based multi-agent systems can automate these processes, their reliability decreases as workflow complexity increases. This project develops a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows and evaluates its performance using both conventional metrics (task success, tool-calling accuracy, latency, token usage, and computational cost) and geometric analysis of latent representations to understand system reliability.

Scope Boundaries

In Scope:

- Develop and evaluate a Recursive Multi-Agent System for enterprise workflows.
- Analyze performance metrics and geometric representations under varying workflow complexity.

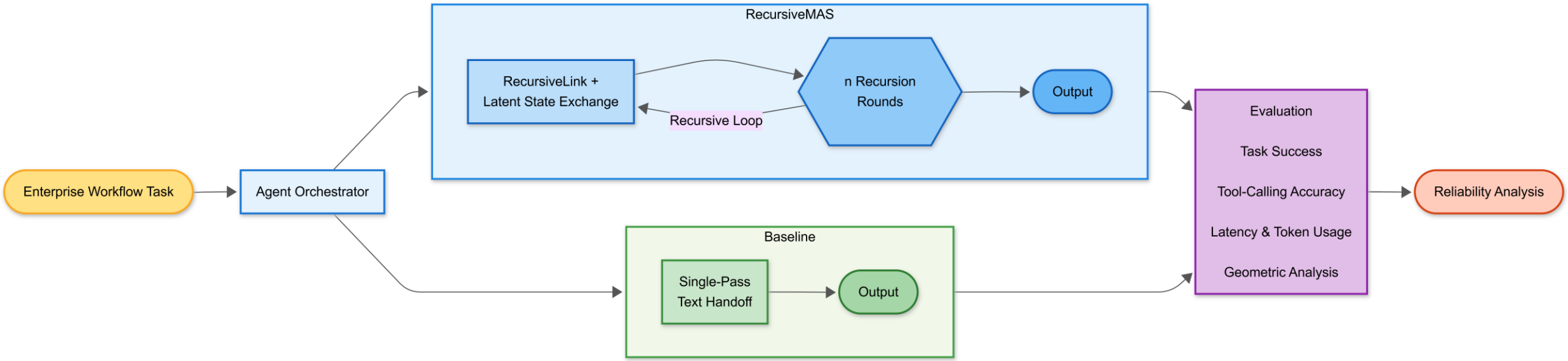
Out of Scope:

- Training or developing new foundation LLMs.
- Production-scale deployment or domain-specific workflow optimization.

Objectives

- Develop a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows with specialized agents (Planner, Tool-Caller, and Verifier).
- Evaluate system performance using task success rate, tool-calling accuracy, latency, token usage, and computational cost across varying workflow complexities.
- Compare recursive and non-recursive (text-mediated) multi-agent systems to evaluate the impact of recursive collaboration.
- Analyze latent agent representations using geometric metrics (e.g., PCA, cosine similarity, representation drift, and effective rank) to study their relationship with workflow complexity and system reliability.

Proposed Methodology / Framework



Techniques & Tools

Component	Tool / Technique
Programming	Python
Agent Framework	LangGraph
Recursive Architecture	**RecursiveLink** – Lightweight 2-layer residual connector for recursive latent-state exchange.
LLM Platform	Hugging Face Transformers / Ollama
Data Processing	NumPy, Pandas
Geometric Analysis	Scikit-learn (PCA), SciPy
Visualization	Matplotlib

Expected Outcomes

- A prototype Recursive Multi-Agent System (RecursiveMAS) for enterprise workflow automation.
- Comprehensive performance evaluation using task success rate, tool-calling accuracy, latency, token usage, and computational cost.
- Geometric analysis of latent representations to study the impact of workflow complexity and recursion depth on system reliability.
- Comparative analysis between recursive and baseline approaches to assess the effectiveness of recursive collaboration.

Timeline

Stage	Deliverable	Jul	Aug	Sep	Oct
1	Framework setup & literature review				
2	Non-recursive baseline implementation				
3	Recursive loop implementation & integration				
4	Comparative performance evaluation				
5	Latent extraction & geometric analysis				
6	Results, report & final presentation				

Team Roles

Team Member	Module / Responsibility
Harshvardhan V	Will develop the recursive communication mechanism (RecursiveLink, connector module, and recursion loop) and perform geometric analysis of latent representations.
Rishi Raajha A	Will conduct performance evaluation, including baseline comparison, task accuracy, latency, computational overhead, and convergence analysis.
Naveen Sankar R S	Will integrate the recursive loop into the framework and perform latent-state extraction and representational similarity analysis.
Sakthi Pranav S	Will carry out the literature review, identify the research gap, and handle documentation and result consolidation.

References

[1] J. Zou, R. Pan, R. Qiu, P. Lu, S. Diao, J. Jiang, H. Tong, T. Zhang, M. J. Buehler, J. He, and J. Zou, “Recursive Multi-Agent Systems”, arXiv preprint, 2026.

[2] Y. Li, Y. Yang, G. Schiebinger, D. D. Cox, and B. A. Olshausen, "Neural Representational Geometry of Concepts in Large Language Models," arXiv preprint, 2024